Activity 5 Guide

Course: Semiconductor Process Fundamentals Masterclass

 Course code: C716
 Package: v1.0 | 1 October 2026
 Activity: 5 - Review an STI Integration Flow
 Alignment: LO2
 Duration: 60 minutes

Goal

Produce the process-flow and challenge analysis for a shallow trench isolation integration.

Scenario

TechSemicon is introducing shallow trench isolation for an advanced CMOS product. Electrical spread and active-area CD variation appear after integration, and the review board needs a complete process/evidence chain.

What You Need

- STI process cards, worksheet and evidence template
- Legacy LOCOS/STI, lithography, etch, CVD and CMP slides
- A3 paper or digital flow-diagram tool

Detailed Step-by-Step Procedure

1. Arrange the STI cards in order: pad oxide, nitride deposition, lithography, trench etch, liner oxidation, trench fill, CMP and nitride strip.
2. For each step, write the purpose, key input and outgoing measurable characteristic.
3. Add pre-clean and post-CMP clean controls at the correct points.
4. Mark the interfaces where active-area CD, trench depth/profile, voiding, dishing/erosion and stress can be introduced.
5. Connect each risk to one suitable measurement such as CD-SEM, cross-section SEM/TEM, film thickness, topography or electrical isolation test.
6. Compare STI with LOCOS and state why STI supports scaling despite its integration challenges.
7. Choose two control points for a pilot-line hold/release plan.
8. Draw the final simplified flow without referring to the cards.
9. Use the checklist to review completeness, sequence and evidence traceability.

Acceptance Criteria

[ ] The eight core STI stages are in the correct order and have a purpose.

[ ] At least four implementation challenges are linked to measurements.

[ ] The flow explains both scaling benefit and integration risk.

Reflection Questions

1. Which STI step most directly controls active-area CD?
2. How can CMP-induced topography alter a later lithography process window?

Evidence to Submit

- Completed worksheet.
- Completed evidence template with file names or screenshots.
- The activity output named with the A05- prefix.
- A final release/hold/recommendation statement supported by measurements.

Troubleshooting

- If a workbook opens in protected view, save a local copy before editing blue input cells.
- If formulas do not update, enable automatic calculation and press Ctrl+Shift+F9 (Windows) or Command+= (macOS Excel).
- If a diagram becomes crowded, separate the process flow from the evidence/control loop; cross-reference them with numbered hand-offs.
- If the conclusion is uncertain, state what is known, what is not known, and the next measurement that can resolve it.